Collection of Documents on Energy Industry

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1. User information needs

The users, who require information on collection of articles in energy industry, are employees and professionals from energy companies, research institutes, Department of Energy in Government, professors and students in university, editors and reporters from energy newspaper, some lawyers, and environmentists.

The users' information need focuses on different sections in energy area, depending on user's identity, requirements, and the area they work. The most required information for them is energy reserves in different forms, such as coal, nuclear, oil, natural gas, and etc. The second, they care about the energy distribution and related affection on economics, whatever in the United States or global scope. The third, they care about the energy history, putting emphasis on the relationship between energy and human survival. The fourth, some professionals care about the energy policy. The fifth, they not only care about the energy production information, but also concern the energy transportation. Later ones are much more important to ensure the energy safety under the situation of unbalanced energy distribution. The sixth, they care about the information related with energy security and corresponding influence on politics and economies in all countries. The seventh, facing the global energy deficiency, they care about some new energy forms, which will play an important roles as the substitute of nowadays energy. The eighth, they care about some new technology adoption on energy exploration and production. The ninth, some companies pay attention to the information of market risk on energy, such as investment risk, profit risk, politics risk, and so on. The tenth, many energy scientists care about the information about different energy forms from research and study vision. The

eleventh, many professionals pay attention to the published information by different energy organizations, such as OPEC, IEA, and NEA.

The user's information needs definitely affect the high level design of the index. For the reason of their concern about energy distribution, the high level index of "district" is appropriate. For the reason of their concern about influence of policies in energy area, high level index of "energy policy" are appropriate. For the reason of their concern about energy reserves by different form, the high level index of "energy reserves" is suitable. There are so many examples that it is hard to list them one by one. In conclusion, the information needs of users definitely affect high level design of index.

2. Subject scope

2.1. Overview

The topics, which are presented in the index on energy documents, focus on energy area. It includes topics, such as energy reserves by different form, energy distribution, energy history, energy policy, energy transportation, energy security, energy substitution, energy technique, energy risk, energy research, and energy organizations.

2.2. Entities

The subject scope of index mainly involves the materials on the area of energy. From the vision of discipline, it mainly involves the energy studies. From the vision of parts and properties, it involves the energy in different forms, such as coal, oil, natural gas, and nuclear energy. From the vision of ideas, it involves the energy policies. From the vision of operations, it involves the energy transportation and exploration. From the vision of process, it involves the energy technique and energy substitution. From the vision of events, it involves the energy securities. From the vision of people, it involves

energy organizations. From the vision of place, it involves the energy distribution in different districts. From the vision of times, it involves the energy history.

The ten examples of high-level categories and related index terms are listed in appendix.

2.3. Assessment

The high level categories are definitely descriptive of the content on energy database. For example, on the aspect of new energy form, such as wind power, tide energy, and solar energy, high level category “energy substitution” is a perfect representation of the content in this energy aspect.

In normal, documentary units in energy collection should be mutually exclusive. However, for the convenience of user searching, some collection parts are aggregated and have a little inevitable overlap. Under some situation, the absolute mutual exclusion sometimes may destroy the consistence and continuation of collections. Hence, in satisfying users need, some collections are overlapped. For example, the documentary unit, which highlights on American energy development, can be found not only under the “District” category, but also under the “Energy history” category. This kind of information organization, even a little overlap, can definitely help users find the aggregated information.

Except the situation talked above, the different classes are in coincidence with the rules of mutual exclusion, and the difference among them are clear and easily anticipated for users. For example, the distinctions among energy reserves, energy substitution, and energy policy classes, are very clear and easily identified.

The collections of documents on energy provide two kinds of access, which separately are controlled vocabulary and free-text keywords. The controlled vocabulary, such as “solar”, “mining”, and “nuclear fission” are specific and easily identified by professional users. Hence, for this part of users, the provision of controlled words is beneficial to improve the search efficiency and usability of the index. However, for the common users who can not accurately describe their information needs or not familiar with energy terms, the free-text keywords are a good substitute. However, sometimes, the controlled vocabulary and free-text keywords are good substitution for each other. As for the collection on energy, indexer mainly organizes the index by using controlled vocabulary. However, it also provides the free-text searching functions for users. This combination way definitely increases the usability of the index for whatever professionals or common users.

From the vision of maintainability, because there is a clear definition for every categories scope and content area, it is definitely easier for indexers to assign documentary units to appropriate categories. For example, a document named “The Biofuels Market Outlook” can be easily classified under “Energy Substitution” category.

3. collection scope

3.1. Creator

There are two kinds of creators, who separately are document units writers and collection creators. The document unit writers include editors and authors of some famous periodical, the freelancers of energy industry, and the researchers of many famous research institutes. However, the collection creators are indexers and professionals, in charge of the collection.

3.2. Title

The tile is “Collection of documents on the energy industry”.

3.3. Format

The format predominantly consists of HTML webpage, PDF file, and partly DOC file. However, a little part of them still adopt printing format.

3.4. Medium

The Medium which represent energy documents, are mainly Webpage.

3.5. Periodicity

Most of contents are published and refreshed by week or month periodically. However for some conference material or books in energy area, they have no fixed periodicity.

3.6. Time and place of publication

The time published is usually in a fixed date of every week, or every month, depending on different kinds of journals. However, for some energy books, the publication time is not fixed. The places of publication are mainly located in the United States, Canada, Russia, Australia, South Africa, Japan, Europe, and China.

3.7. Rights

The rights belong to the author or periodical publisher.

3.8. Level (audience)

The audience mostly consists of managers or directors who are responsible for design-making in energy companies, some policy-maker from government department, researchers from University, some lawyers, and environmentists.

3.9. Form

It mainly consists of articles form. A little part of them is books and conference materials.

3.10. Style

The style is very serious-minded, thoughtful, meditative, and reflective.

3.11. Genre

The genre is analytical and current.

3.12. Domain

The domain focused on energy.

3.13. Language

The languages used are chiefly English, Japanese, Russian, and Chinese.

3.14. Symbols

It includes alphabetic code, digit, and some illustration chart.

3.15. Assessment of impact

The clarified collection scope of energy is helpful in paving road on building index. Because it primarily consist of document in electric form, it is relatively easy for maintaining.

4. Collection (documentary) domain

4.1. The sources

The documents are extracted from periodicals (Such as COAL, Oil & Gas Journal, and Energy Journal) , research institutions (Such as Global Business Insights), and some publishers, highlight on energy area, such as coal, oil, natural gas, electricity, and nuclear power.

4.2. Secondary sources

For secondary source being used, the most obvious characteristics are “Reserves”, “District”, “History”, “Policy”, “Substitution”, “Energy Transportation”, “Energy Technique”, “Energy Risk”, “Energy forms”, “Energy Safety”, and etc.

4.3. Assessment of impact

The collection domain of energy has complicated characteristics that need the index on catalog and classification, in helping users to find the target easily. As for the control vocabulary on secondary index, it has a little inconvenience in updating them and maintaining consistency. Furthermore, the people in charge of the collection need to make sure those new articles, journals, and reports are added into the collection in time.

5. Display medium of the index

5.1. Medium of representation

The medium used to display the index is web page, which consist of XML, HTML, CSS functions. These kinds of medium are convenient for user to search. Because the web page have good share ability, it is convenient for many users to share the recourses simultaneously. The Search engine attached on the webpage provides a good channel for the information retrieval of users.

5.2. Code and symbols

The index mainly uses English letters, digits, and punctuation marks, abbreviation to represent the content of it. For example, it uses English letters to demonstrate meaningful English words or phrase, use organized digit to represent the date.

6. Indexable matter

The indexes are extracted from the main aspects or representation among articles, conference materials, and books in energy area.

For example, “Energy Reserves” is the important portion of database. It represents the information on energy reserves in different fields, such as coal, oil, natural gas, and etc. The purpose is to give the users or professionals, who care about energy reserves, an easily searching way. Moreover, some search and sorting application can help to reach the target of easy-use. By using modern information technology, such kinds of index have high usability for most of users. Furthermore, because of the clear boundary among different collections, it is very convenient on maintainability of index.

7. Displayed and non-displayed indexes

7.1. Index display form

The indexes are partly displayed for the convenience of users’ browsing material in a clearly organized way. If stacking all of indexes together, it will puzzle the users. Hence, only important indexes, which play the role of usher, are displayed. Many none-displayed indexes adopt the form of free-text, hidden on the background. However, it is very convenient to find the information within none-displayed indexes by using free-text searching.

7.2. Affection on usability

The index on energy documents is in electronic form, promoting the usability, and facilitating the browsing very much. The reason is as below. The first, it provides share-browsing mode, making many users can browse the same document simultaneously. The second, electronic format indexes are easily to adjust or classified. It can be changed reasonably by adopting the advice coming from professional or patrons. The third, the combination of displayed and non-displayed index, can help to satisfy information needs

of users, who have different knowledge background. Certainly, it increases the usefulness and usability of indexes.

8. Classified/relational arrangement

8.1. Classifies or Relational index

The index on energy documents adopts a classified and relational arrangement, which demonstrate a semantically consistent relationship between super categories and subcategories. Two kinds of relationships are adopted, which separately are whole-part and entity-process relationships. For example, as for the preceding one, the relationship between super category “energy reserves” and subcategories, such as “coal reserve”, “oil reserve”, are definitely whole-part relationships. As for the later one, the relationship between super category “Energy transportation” and subcategories, such as “Line transportation”, “pipe transportation”, are definitely entity-process relationships.

Furthermore, these relationships are clear and predictable from semantic vision. Moreover, the categories at the same level are distinctive and mutual exclusive. For example, under “Energy reserves” categories, the subcategories of “oil”, “natural gas”, and “coal” are at the same level, simultaneously clearly distinctive and mutual exclusive.

8.2. Facets

The index also uses facets to demonstrate individual aspects of a subject (Cleveland, 2001). The facets, my index in use, are “energy reserves, district, energy history, energy policy, energy transportation, energy security, energy substitution, energy technique, energy risk, energy research, and energy organization”. The facets are organized according to the rank which comes from the statistics on user search frequency. In other words, they represent the order of facets in which users like to use.

8.3. Usage of classified index

The questions, which are fit to be answered in classified arrangement form, are semantically consistent and clearly organized ones. However, some vague and inconsistent questions are not suitable for classified or relational arrangement.

9. Vocabulary management

9.1. Controlled vocabulary

The index on energy documents adopt the controlled vocabulary, which are extracted from meaningful words or syntactic forms. There is no existing resources or standard used for vocabulary control. Hence, controlled vocabulary must be extracted depending on the index author's abundant experience and deep understanding in energy area.

9.2. The resources for vocabulary control

The resources for vocabulary control come from the thesaurus list, from indexer's knowledge accumulation.

9.3. Synonymous terms

The synonymous and equivalent terms are linked in a thesaurus, which, on the one hand, used as the reference to strengthen index, one the other hand, employed as the a way to strengthen search applications. "Conversion, *see also* transmission" is a good example (see fourth page on Index Design).

9.4. Homography

The homograph must be distinguished by injecting some notes in brackets following them. For example, the word "gas" in energy usually has many kinds of

meaning. The first is natural gas, the second is coal gas. They are distinguished by adding bracket method, such as “gas (natural)” and “gas (coal).

9.5. Semantic specificity

Broad term and narrower terms are delicately controlled in the energy index. For example, “Crisis” is a broad term, and “1973 crisis” is the narrow term which restricted the scope by year.

9.6. Other conceptual relations

The other conceptual relations, used in index, are name relations control. The same person will have different name in his/her whole life, such as Marie Curie, who has another name Marie Skłodowska-Curie. Different name of the same person must be demonstrated. These kinds of name relations control will not only improve the recall, but also improve the precision.

9.7. Impact on usability

Vocabulary management policy definitely increases the usability of index, on the aspects, such as improving precision, improving recall, improving search efficiency, and giving users convenience in using index.

10. Analysis and indexing method

10.1. Text representation

The information of content, purpose, meaning, features and potential use of a text, are obtained by make a deep analysis to the users requirement. Sometime, visiting patrons or users is an indispensable way. For example, after talking with users in energy area, their requirements are clarified. Through analysis, many aspects on energy industry, such as what is the future of it, what is the history of it, what is the component of it, what

is the symptom of it, what is other usage of it, and what is the affection for economic development, are clarified. At the same time, some questionnaire assisted by computer can play an important role in the analysis process. Some computer algorithms can help analyzer to save lots of strength. However, finally, it is the human, not computer, to handle the essence or spirit of whole index process and make representation.

10.2. The balance between human index and computer index

At first, the common terms are extracted by human intellect or by computers with human-designed algorithms. Then, human indexer selects the representative terms or descriptors from them according semantic connection between different concepts and information organization requirements.

The computer indexing definitely can be used to improve handling efficiency on pre-indexing phase. Moreover, it can be used as a faster way to extract terms from documents for the use of free-text searching. However, in the area of complicated category or subject heading indexing, it must depend on the human intellect. Hence, it is definitely a balanced way, to combine the human indexing and computer indexing together, during index process. In this way, it not only conquers the flaw of computer index, but also improves the human indexing efficiency.

10.3. Usability and maintainability

Combining usage of human index and computer index shrinks time index made, hence it improves the usability of the index. The mixture attitude improves the efficiency of index maintenance under the help of computers; simultaneously keep the high precision level under the human intellect.

11. Exhaustivity of indexing

11.1. The identification of exhaustivity

There is a high exhaustivity on the index of energy documents, because average number of descriptors, assigned to documents, reaches from 7 words per document unit.

11.2. Example

The exemplar document is as below. For the reason of space limit, only the title is displayed, which is “The Future of Clean coal: The impact of new technologies and legislations on the economics of coal-fired power generation” (Breeze, 2007).

11.3. Retrieval access point

The total retrieval access points for this document are listed below, such as “coal”, “clean”, “technology”, “policy”, “economics”, and “cost”. For each access points, the main heading or subheading are list below.

11.4. Complete heading list on access point

Order	Access point	Main heading	Subheading
1	Coal	Energy Reserves	Coal
2	Clean	Energy Technique	Clean Energy Technique
3	Technology	Energy Technique	Clean Energy Technique
4	Policy	Energy Policy	Pollution-Prevention Policy
5	Economics	Energy Security	Economics
6	Cost	Energy Risk	Cost Risk

11.5. The number of descriptors

The index on energy documents provides two kinds of indexes, which separately are displayed and none-displayed index. From the vision of displayed index, the total number of unique descriptors, assigned to the document, reaches to nearly 7 words per

document unit. From the vision of non-displayed index, the stemming method is used to simplify the search process, and improve the efficiency in full-text search. In order to increase the recall, stop words scope are shrunk as small as possible.

11.6. Relationships between exhaustivity and precision and recall

The higher the exhaustivity is, the higher the recall is. However, sometimes, high exhaustivity may result in the low precision because “not only highly relevant documents but also marginally relevant ones will be retrieved” (Wellisch, 1995). For example, under the low exhaustivity situation, when a user want the nuclear reserves distribution, he input a search term “nuclear”, and get one results he required. However, under the high exhaustivity situation, when he input a search term “nuclear”, he not only get the results of “nuclear reserves” which is his target, but also get the results of “nuclear fusion” which is not his target.

11.7. The usability and maintainability

The exhaustivity of index on energy document is very high. It improves the usability of index, and gives a greater probability for users to find index terms which are relevant. However, there are also some deficiencies. The first, user must browse through more index term before finding target. The second, user has to spend lots of time in search because of disturbance coming from the likelihood on a greater number of irrelevant items.

High exhaustivity of index on energy document, means indexer has to spend much more time to extract more terms from documents. This definitively increases the difficulties of index creation and maintainability.

12. Specificity of indexing terms

12.1. The policy relating with specificity

The policy, regarding specificity for the index on energy document, is to use different specificity for different subject. Because some subject in energy area does not need high specificity, in this situation, low specificity strategy is employed. However, in some situation which requires high specificity, high specificity must be used. Whether to use high or low indexing specificity depends on the concrete situation and subject characteristics. However, once the policy is made, it must be abided by to keep consistence.

12.2. Thematic specificity

For example, as for the “energy technique” topic, because users or patrons require a good guide to dive into the appropriate subcategory or contents, high thematic specificity are provided to satisfy their requirements. However, as for the “energy reserve” topic, users or patrons can find required information for its relatively simple characteristics. In this situation, low specificity is enough to satisfy the user needs.

Another example, when patrons talk of coal “mining technology”, they immediately dive into the mining method topic. Hence, it is suitable for the index to provide detailed subthemes such as “longwall mining” or “shortwall mining” with high specificity, helpful for users to dive into the appropriate material immediately.

12.3. Operational specificity

As for the operational specificity of the index, in some subject such as “mining technology”, the indexer allocates fewer documents to a term, keeping a higher operational specificity in helping users find appropriate document unit. In other subject,

such as “energy reserves”, it allocates more documents to a term, keeping a lower operational specificity to help users locate the suitable categories.

12.4. Usability and maintainability

The high thematic specificity of the index has a good usability for users, for its providing good way to help them find appropriate documents. However, this kind of high specificity sometimes easily leaks some important aspects, resulting low recall. The situation of low thematic specificity is reverse. The high thematic specificity of the index requires indexer to pay much more energy on allocating peculiar descriptors on document units, so the maintainability of the index is not good. The situation of low thematic specificity on maintainability is reverse.

13. Syntax

13.1. Identifying the syntax

The index on energy document has two parts which separately are displayed index and non-displayed index. As for the displayed index, the syntax used to represent headings, are mainly natural language syntax, using standard word order, using hyphen between years, using title case or capitalization. The displayed index is precoordinate index.

As for the non-displayed index, the syntax, which used to combine terms in search statements, is Boolean combination, bracket marks, plus quotation marks, searchable categories choice, and advanced search functions (see page 5 on Index Design). The non-displayed index is postcoordinate index.

13.2. Ten index heading examples

Order	Heading example	Syntax
1	Geothermal Power	Natural language order
2	17-19 Century	Punctuation-hyphen
3	Turbine Technique	Title case and Natural language order
4	Energy Transmission	Title case and Natural language order
5	Technique Risk	Title case and Natural language order
6	Radiant Energy	Title case and Natural language order
7	International Atomic Energy Agency	Title case and Natural language order
8	Natural Gas Hydrate	Title case and Natural language order
9	Energy Safety/Security	Punctuation-slash
10	Pollution-Prevention Policy	Punctuation-hyphen

13.3. Usability

The syntax on searching and retrieval is very important. For example, natural order syntax is easily understood by users during searching or browsing. The “Boolean operator” and “searchable categories choice” on non-displayed index, has great usability on searching, providing great convenience for information retrieval. Simultaneously, it improves the search efficiency.

14. Alphanumeric arrangement

14.1. Whether take precedence

If the displayed indexes are alphabetical, concept or symbol should abide by the order allocated to them in alphabetization regulations. It will help to keep consistence. However, for non-alphabetical index, concept should be viewed as important, and taking precedence.

14.2. Illustration examples

The example illustrating the arrangement of digits, symbols, and alphabetical characters can see the page 4 on Index Design.

14.3. Electronic media

The index on energy documents do not used the electronic multimedia. If they are used, the metadata is a good choice as an identification or foundation to display, which includes multimedia capacity, publication time, the content schema, and so on.

14.4. Example

Alphabetical index example demonstrated on the page 4 on Index Design. The classified and relational index example demonstrated under caption 8 which is classified/relational arrangement. The facet cites order is identified in that section. Indeterminate or associate index are also listed under caption 8.

15. Locator and links to documents

15.1. The links connecting index heading and documents

In the index on energy documents, hyperlink is used to connect index headings or terms to documents. These hyperlinks hide behind the index headings, hence invisible. When users click the index heading, hyperlink takes effect and drives the user to a required information page.

15.2. Internal locators

The internal or intra-database hyperlinks can lead users to full text segments within the database. It is very helpful for users to locate the information inside a big full-text document. Linking to surrogate is another useful functions which can help users scan the metadata of documents in a fast way, very helpful to increase the search speed.

15.3. The steps to find document surrogates

In order to find document surrogates or full text document, information seeker can have two choices. The first choice, they must follow the directions on index website,

selecting the corresponding subject heading, and then entering into corresponding web page through hyperlinks. The second choice, they can directly find document surrogates through searching non-displayed index. Then they can find the full text document through the links in surrogates.

16. Search interface

16.1. Opening screen

The index on energy document is electronic databases. The opening screen makes a clear description for the subject domain, collection domain, and search options (see page 1 on Index Design). There are some top buttons, such as “Subject browsing” and “Title browsing”, providing a good guidance for users into corresponding pages. There are 4 picture button, clearly demonstrating and guiding users into the main energy areas, such as “coal”, “oil”, “electricity”, and “natural gas”. The highest levels of classified index are marked with underline, telling users some hyperlinks hidden behind them, and guiding users to visit them by clicks.

16.2. The presentation of displayed indexes

The displayed indexes are presented to user by the form of organized subject. They are presented in different stages, very convenient for user’s browsing. Furthermore, the interface designer must concern some questions, such as the information capacity of every entry, the methods to connect the index. The non-displayed indexes can be presented to users through advanced search interface, by using the method of machine-term-matching (see page 5 on Index Design). The effectiveness of displays is decided by the factors, such as the reasonable layout of WebPages, coincidence to user browsing habit.

16.3. Search syntax

The search syntax is the methods by which search elements are combined. Search elements includes “Boolean”, “quotation marks”, “bracket marks”, ways to broaden or limit search, and searchable categories choice. The search syntax is clearly clarified on Index Design (see page 5 on Index Design).

17. Surrogate displays

17.1. Definition of surrogate

The data about the individual document and metadata, within the index on energy documents, are bibliographical data, index terms, subject headings, descriptors, locators, information about size, availability, quality, abstract, extract of context, and abstract, etc.

17.2. Forms of surrogate data

The index on energy documents provides two kinds of surrogate data. One of them is a short surrogate data for the convenience of seeker’s fast browsing (see Page 6 on Index Design). Another of them is a unified display of all surrogate data (see Page 7 on Index Design).

17.3. Surrogate displays

The index provides a surrogate display options button, named “Full Surrogate Data Display”, located on page 6 of Index Design. If a seeker wants to browse the full surrogate data, she can enter into the page named “full records surrogate” by clicking this button. The “index records” page contains a link to browse complete documents, so it is actually a full-text database.

The elements in surrogates encompass “Personal author”, “Title”, “Source”, Publication year”, “Subject”, “Abstract”, “Location”, “Publication Info”, and so on (See Page 7 on Index Design).

17.4. Surrogates display example

The search term, “Energy Crisis”, is put into search box. As a result, there are two records returned from internal search engine to users. The first one of them is “The final energy crisis”. The second one of them is “understands the 1990 oil crisis” (See Page 6 on Index Design).

17.5. Impact on retrieval

The surrogate size can be different. With the short surrogate, users or patrons can scan more records faster. With the long surrogate, users can know much more information about the document unit. However, for laymen or novice, the long surrogates sometimes are too complicated to understand.

18. Conclusion

In conclusion, organizing information is very important, not only for libraries, but also for human civilization promotion. Simultaneously, information organizations are in continuous evolution with the development of technology and World-Wide Web. Human will always play leading roles in the aspect of organizing information.

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